B.Sc. (Hons.) Agriculture - First Semester · Model Question Paper - 1 (50 marks pattern)

Agricultural Heritage

(कृषि विरासत)

AGR-102 · Credit 1 (1+0)

Model Paper 1 with Solutions (important semester questions)

मॉडल पेपर 1 - सेमेस्टर के लिए महत्वपूर्ण प्रश्न, हल सहित

Time	Maximum Marks	Sections
2 Hours	50	A: 5 × 2 = 10 · B: 4 × 5 = 20 · C: 2 × 10 = 20

Not an official university paper; prepared on the syllabus and usual semester pattern. (विश्वविद्यालय का आधिकारिक पेपर नहीं)

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SECTION A

Very Short Answer Type Questions

अति लघु उत्तरीय प्रश्न

Attempt all five parts of Q.1. Each part carries 2 marks; maximum 50 words.

प्रश्न 1 के सभी पाँच भाग अनिवार्य। प्रत्येक 2 अंक; अधिकतम 50 शब्द।

5 × 2 = 10 Marks

Q.1 (a) What is Krishi-Parashara? Mention its main contents.

2 Marks

कृषि-पराशर क्या है? इसकी मुख्य विषय-वस्तु बताइए।

Krishi-Parashara (कृषि-पराशर)

An ancient Sanskrit text on agriculture attributed to sage Parashara (dated between about 400 and 1100 AD by different scholars). It treats farming as the root of wealth and gives practical rules for the farmer.

महर्षि पराशर रचित कृषि का प्राचीन संस्कृत ग्रन्थ; कृषि को धन का मूल मानता है।

Main contents (मुख्य विषय)

- Rain forecasting by planets and clouds
  वर्षा पूर्वानुमान
- Care and management of cattle
  पशु प्रबन्धन
- Plough and its parts; ploughing time
  हल व जुताई
- Seed collection, storage, sowing
  बीज संग्रह व बुवाई
- Weeding, harvesting, threshing, manure (gomaya)
  निराई, कटाई, गोबर खाद

Q.1 (b) Who wrote Vrikshayurveda? Write its importance.

2 Marks

वृक्षायुर्वेद किसने लिखा? इसका महत्त्व लिखिए।

Vrikshayurveda (वृक्षायुर्वेद)

"Science of plant life" written by Surapala (about 1000 AD). It deals with tree planting, nursery, irrigation, nourishment and treatment of plant diseases, like Ayurveda does for humans.

सुरपाल (लगभग 1000 ई.) द्वारा रचित "पौधों के जीवन का विज्ञान" - पौध रोपण, सिंचाई, पोषण व पौध रोग उपचार।

Importance today (आज का महत्त्व)

- Basis of herbal plant protection
  हर्बल पौध संरक्षण
- Gives Kunapajala liquid manure
  कुणपजल तरल खाद
- Guides orchard and garden layout
  बाग नियोजन
- Supports natural / organic farming
  प्राकृतिक खेती

Note: Varahamihira's *Brihat Samhita* (6th century AD) also has a chapter on Vrikshayurveda.

वराहमिहिर की बृहत्संहिता (6वीं सदी) में भी वृक्षायुर्वेद का अध्याय है।

Q.1 (c) What is Kunapajala?

2 Marks

कुणपजल क्या है?

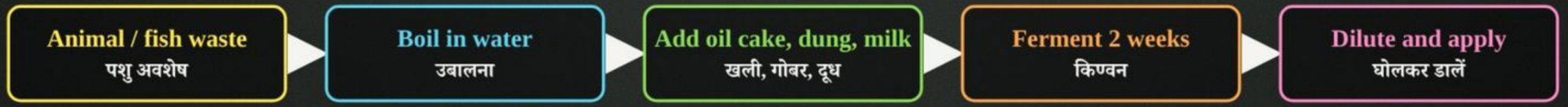
Kunapajala (कुणपजल)

A traditional fermented liquid manure described in Vrikshayurveda, prepared by boiling animal waste (flesh, fat, bone marrow of animals or fish) and fermenting it with water, oil cake, dung, urine, milk and honey.

वृक्षायुर्वेद में वर्णित किण्वित तरल खाद - पशु / मछली अवशेष को उबालकर जल, खली, गोबर, गोमूत्र, दूध व शहद के साथ सड़ाया जाता है।

Uses (उपयोग)

- Plant growth promoter
  वृद्धि वर्धक
- Supplies N, P, K and microbes
  पोषक तत्व व सूक्ष्मजीव
- Vegetarian "herbal kunapajala" now made
  हर्बल कुणपजल



Q.1 (d) **Name any four agricultural revolutions of India with their products.** **2 Marks**

भारत की किन्हीं चार कृषि क्रान्तियों के नाम व उनसे जुड़े उत्पाद बताइए।

Revolution	Product	Key person / year
Green (हरित)	Food grains - wheat, rice	M.S. Swaminathan, 1966 - 67
White (श्वेत)	Milk	Verghese Kurien - Operation Flood, 1970
Blue (नीली)	Fish	Hiralal Chaudhuri and Arun Krishnan
Yellow (पीली)	Oilseeds	Sam Pitroda - Technology Mission on Oilseeds, 1986
Golden (स्वर्णिम)	Horticulture / honey	Nirpakh Tutej (horticulture)

Q.1 (e) Who were Ghagh and Bhaddari? Give their contribution. — 2 Marks

घाघ और भड्डुरी कौन थे? उनका योगदान बताइए।

Ghagh and Bhaddari (घाघ व भड्डुरी)

Famous folk poets of north India whose short sayings (kahavat) in Awadhi / Bhojpuri give farmers rules on weather forecasting, sowing time, soil, cattle and crops. They form a part of our indigenous knowledge.

उत्तर भारत के लोक कवि; उनकी कहावतें मौसम, बुवाई, मिट्टी, पशु व फसल के बारे में किसानों को नियम बताती हैं।

Example saying (उदाहरण)

"शुक्रवार की बादरी, रहे शनीचर छाय। तो यों भाखै भड्डुरी, बिन बरसे ना जाय॥"

Clouds on Friday that stay till Saturday surely bring rain.

Importance: such observations are now studied as traditional weather indicators alongside IMD forecasts.

महत्त्व: अब इन्हें IMD पूर्वानुमान के साथ परम्परागत मौसम संकेतक के रूप में अध्ययन किया जाता है।

SECTION B

Short Answer Type Questions

लघु उत्तरीय प्रश्न

Attempt any four out of six questions. Each carries 5 marks; maximum 100 words. (All six solved here)

छः में से किन्हीं चार प्रश्नों के उत्तर दीजिए; प्रत्येक 5 अंक। (यहाँ सभी हल)

4 × 5 = 20 Marks

Q.2 Describe agriculture in the Indus Valley (Harappan) civilisation. 5 Marks

सिन्धु घाटी (हड़प्पा) सभ्यता में कृषि का वर्णन कीजिए।

The Harappan civilisation (about 2600 - 1900 BC, mature phase) was based on surplus farming on fertile flood plains of the Indus. Earliest farming of the subcontinent is from Mehrgarh (about 7000 BC).

हड़प्पा सभ्यता (लगभग 2600 - 1900 ई.पू.) सिन्धु के उपजाऊ बाढ़ मैदानों में अधिशेष खेती पर आधारित थी; सबसे प्राचीन खेती मेहरगढ़ (7000 ई.पू.)।

Feature	Evidence
Crops	Wheat, barley, peas, sesame, mustard; rice at Lothal and Rangpur
Cotton	Earliest cotton cultivation in the world (cloth piece at Mohenjo-daro)
Ploughed field	Kalibangan (Rajasthan) - criss-cross furrows, sign of mixed cropping
Storage	Large granaries at Harappa and Mohenjo-daro
Animals	Humped bull, buffalo, sheep, goat domesticated
Irrigation and trade	Flood water, wells; Lothal dockyard for trade of farm produce

Q.3 Write a short note on agriculture in Kautilya's Arthashastra. 5 Marks

कौटिल्य के अर्थशास्त्र में कृषि पर संक्षिप्त टिप्पणी लिखिए।

Arthashastra (Kautilya / Chanakya, about 300 BC, Mauryan period) is a treatise on statecraft with detailed rules for agriculture.

अर्थशास्त्र (कौटिल्य, लगभग 300 ई.पू., मौर्य काल) - राज्य व्यवस्था का ग्रन्थ जिसमें कृषि के विस्तृत नियम हैं।

Topic	What Arthashastra says
Sitadhyaksha	Superintendent of agriculture who managed crown lands (sita)
Rain gauge	*Varshamana* - rainfall measured; sowing planned on rainfall
Seed treatment	Seeds exposed to dew and heat, treated with honey, ghee, cow dung
Irrigation tax	*Udakabhaga* - one-fifth to one-third of produce for state-built irrigation
Land revenue	Usually one-sixth (*bhaga*) of produce
Crop planning	Rice and early crops first, later crops by rain; protection of cattle and pastures

Q.4 Discuss the relevance of agricultural heritage to present-day agriculture. 5 Marks

वर्तमान कृषि में कृषि विरासत की प्रासंगिकता पर चर्चा कीजिए।

Modern intensive farming has raised yields but caused soil, water and climate problems; heritage offers low-cost, sustainable solutions.

आधुनिक सघन खेती ने उपज बढ़ाई पर मृदा, जल व जलवायु समस्याएँ दीं; विरासत सस्ते व टिकाऊ समाधान देती है।

Heritage practice	Present-day relevance
Mixed cropping, crop rotation	Risk reduction, soil fertility, pest break - basis of cropping system research
Cow-based inputs, Kunapajala	Natural farming (Jeevamrut), National Mission on Natural Farming
Rainwater harvesting (tanks, johads)	Watershed programmes, Amrit Sarovar, farm ponds
Native landraces and breeds	Gene source for climate-resilient varieties; GI tags (Kalanamak rice)
ITK pest control (neem)	Neem-coated urea, bio-pesticides, IPM
Millets	Nutri-cereals; International Year of Millets 2023

Conclusion: blending heritage with science gives sustainable and climate-smart agriculture.

निष्कर्ष: विरासत + विज्ञान = टिकाऊ व जलवायु-स्मार्ट कृषि।

Q.5 Explain plant protection through indigenous traditional knowledge.

5 Marks

स्वदेशी परम्परागत ज्ञान द्वारा पौध संरक्षण समझाइए।

Field practices (खेत में उपाय)

- Neem leaf / seed kernel extract spray
  नीम अर्क छिड़काव
- Cow urine + neem / garlic - chilli spray
  गोमूत्र, लहसुन - मिर्च
- Dusting of wood ash on vegetables
  राख का भुरकाव
- Sour buttermilk spray against fungal disease
  खट्टी छाछ
- Bird perches (T-shaped) for insect-eating birds
  पक्षी आश्रय
- Trap crops - marigold, mustard border
  प्रपंच फसल

Storage and seed (भण्डारण व बीज)

- Dry neem leaves mixed in grain bins
  नीम पत्ती
- Mud bins (kothi) sealed with cow dung
  मिट्टी की कोठी
- Pulses coated with mustard oil / ash
  तेल / राख लेप
- Sun drying before storage
  धूप में सुखाना
- Seed treated with cow dung slurry
  गोबर घोल उपचार

Why it works (वैज्ञानिक आधार)

Neem has azadirachtin (antifeedant), ash and oil block insect movement, and birds and trap crops give natural control - low cost and no residue.

नीम में एज़ाडिरेक्टिन; राख व तेल कीट गति रोकते हैं; सस्ता व अवशेष-रहित।

Q.6 Compare the past and present status of agriculture and farmers in society. 5 Marks

समाज में कृषि एवं किसानों की पूर्व व वर्तमान स्थिति की तुलना कीजिए।

Period	Status of farmers	हिन्दी सार
Ancient	Farming respected; self-sufficient village economy; tax about one-sixth	सम्मानित, आत्मनिर्भर गाँव
Medieval	Heavy land revenue (up to one-third); Todar Mal's revenue system; new crops	भारी लगान
British	Zamindari (Permanent Settlement 1793), forced indigo (Champaran 1917), famines, debt	ज़मींदारी, अकाल, शोषण
After 1947	Zamindari abolition, land reforms, cooperatives, Green Revolution, MSP	भूमि सुधार, हरित क्रान्ति
Present	Income support (PM-KISAN), KCC, insurance, FPOs, digital markets; but small holdings and low income	योजनाएँ, पर छोटी जोत

Trend: from respect and self-reliance → exploitation in colonial times → recovery and food surplus today, with the new challenge of raising farm income.

प्रवृत्ति: सम्मान → औपनिवेशिक शोषण → आज खाद्य अधिशेष; अब चुनौती आय बढ़ाने की।

Q.7 Describe the national agricultural setup (research, education and extension) in India. 5 Marks

भारत में राष्ट्रीय कृषि ढाँचे (अनुसंधान, शिक्षा व प्रसार) का वर्णन कीजिए।



Body	Role
ICAR (1929)	Apex research body; over 100 institutes and national research centres
Agricultural universities	About 75 SAUs, central and deemed universities for teaching and research
KVK (first 1974)	More than 730 district centres for on-farm testing and training
ATMA (2005)	District extension through farmer groups
Support bodies	NABARD (credit), CACP (MSP), FCI (procurement), NSC (seed), APEDA (export)

SECTION C

Long Answer Type Questions

दीर्घ उत्तरीय प्रश्न

Attempt any two out of four questions. Each carries 10 marks; maximum 400 words. (All four solved here)

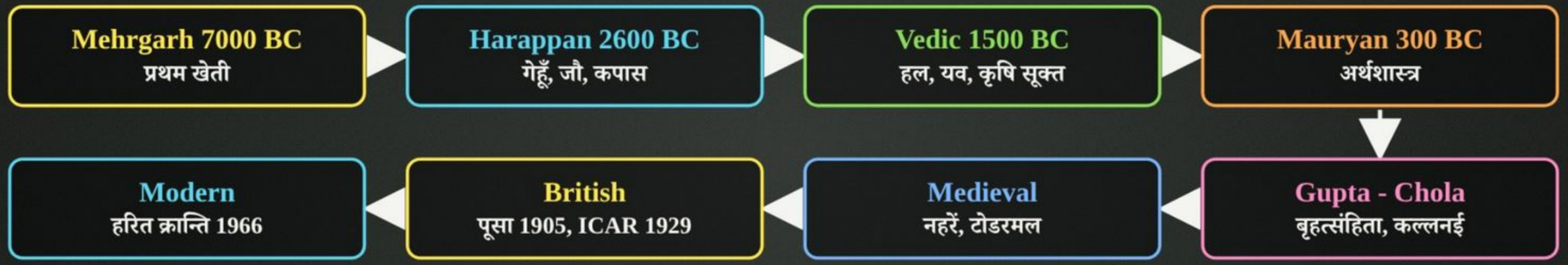
चार में से किन्हीं दो प्रश्नों के उत्तर दीजिए; प्रत्येक 10 अंक। (यहाँ सभी हल)

2 × 10 = 20 Marks

Q.8 Trace the journey of Indian agriculture from the ancient to the modern era.

10 Marks

प्राचीन से आधुनिक युग तक भारतीय कृषि की यात्रा का वर्णन कीजिए।



Era	Key developments
Ancient (to 600 AD)	Domestication of crops and animals; plough; Kalibangan ploughed field; Arthashastra rules; texts on rain and seeds
Early medieval	Krishi-Parashara, Vrikshayurveda (Surapala), Kashyapiya Krishisukti; tank irrigation in south
Sultanate - Mughal	Western Yamuna canal (Firoz Shah Tughlaq), Persian wheel; Todar Mal land revenue; Portuguese bring maize, chilli, potato
British (1757 - 1947)	Commercial crops (indigo, cotton, tea); zamindari; famines; Pusa institute 1905; ICAR 1929

Q.8 Journey after independence and modern era

contd.

स्वतंत्रता के बाद व आधुनिक युग

Year	Milestone	हिन्दी
1950s	Zamindari abolition, Community Development (1952), first Five Year Plan priority to agriculture	भूमि सुधार
1965	Agricultural Prices Commission (now CACP) and Food Corporation of India - MSP and procurement	MSP, FCI
1966 - 67	Green Revolution - HYV wheat and rice	हरित क्रान्ति
1970	Operation Flood - White Revolution	श्वेत क्रान्ति
1974 / 1982	First KVK; NABARD set up	KVK, NABARD
1986	Technology Mission on Oilseeds - Yellow Revolution	पीली क्रान्ति
2000 - 2016	National Agricultural Policy, NFSM (2007), Soil Health Card, PMFBY, e-NAM	नीतियाँ व योजनाएँ
2019 onwards	PM-KISAN, 10,000 FPOs, drones, Digital Agriculture Mission, natural farming mission	डिजिटल व प्राकृतिक खेती

Conclusion (निष्कर्ष)

Indian agriculture moved from subsistence to scarcity under colonial rule and then to food surplus; the present aim is sustainable, profitable and climate-smart farming.

जीविका खेती → औपनिवेशिक कमी → खाद्य अधिशेष; अब लक्ष्य टिकाऊ, लाभकारी व जलवायु-स्मार्ट खेती।

Q.9 Discuss the scope of agriculture in India.

10 Marks

भारत में कृषि के कार्यक्षेत्र (स्कोप) की विवेचना कीजिए।

Scope of agriculture means the range of activities, branches and opportunities covered by agriculture - from producing crops to processing, marketing, research and services.

कृषि का कार्यक्षेत्र = फसल उत्पादन से प्रसंस्करण, विपणन, अनुसंधान व सेवाओं तक की सभी गतिविधियाँ व अवसर।

Branches of agriculture (कृषि की शाखाएँ)

Branch	Deals with
Agronomy	Field crop production
Horticulture	Fruits, vegetables, flowers
Soil science	Soil and nutrients
Plant breeding	New varieties
Plant protection	Pests and diseases
Animal husbandry, dairy	Livestock
Agricultural engineering	Machines, irrigation
Extension, economics	Transfer of technology, markets

Allied sectors (सहायक क्षेत्र)

- Fisheries, poultry, apiculture
  मत्स्य, मुर्गी, मधुमक्खी
- Sericulture, mushroom, floriculture
  रेशम, मशरूम, पुष्प
- Agroforestry, medicinal plants
  कृषि वानिकी
- Food processing, value addition
  खाद्य प्रसंस्करण

Q.9 Opportunities and future scope

contd.

अवसर व भविष्य का कार्यक्षेत्र

Area	Opportunity
Government jobs	Agriculture officer, ADO, technical assistant, scientist (ARS), banking (agriculture field officer)
Agri-business	Seed, fertiliser, pesticide, machinery companies; Agri-Clinics and Agri-Business Centres (2002)
Entrepreneurship	Agri-startups, nurseries, mushroom, vermicompost, dairy units, FPOs
Export	Rice, spices, fruits, organic produce; APEDA support
New technology	Precision farming, drones, protected cultivation, hydroponics, biotechnology
Sustainability	Natural and organic farming, carbon credits, climate-resilient agriculture

Conclusion (निष्कर्ष)

Agriculture is not only farming but a wide agri-food system; with nearly half the workforce and rising demand for food, nutrition and exports, its scope in India is vast and growing.

कृषि केवल खेती नहीं, सम्पूर्ण कृषि-खाद्य प्रणाली है; भारत में इसका कार्यक्षेत्र विशाल व बढ़ता हुआ है।

Q.10 **Describe the important ancient agricultural texts of India and their contribution.** **10 Marks**

भारत के प्रमुख प्राचीन कृषि ग्रन्थों व उनके योगदान का वर्णन कीजिए।

Text	Author / period	Contribution
Rigveda, Atharvaveda	Vedic period (1500 - 600 BC)	Plough, barley, irrigation, prayers for rain, pest charms
Arthashastra	Kautilya, about 300 BC	Agricultural administration, rain gauge, seed treatment, irrigation tax
Brihat Samhita	Varahamihira, 6th century AD	Rain forecasting, Vrikshayurveda chapter, groundwater indicators
Krishi-Parashara	Sage Parashara, about 400 - 1100 AD	Rain prediction, cattle, plough, seed, harvesting, manure
Kashyapiya Krishisukti	Kashyapa, about 800 AD	Rice cultivation, transplanting, land and water management
Vrikshayurveda	Surapala, about 1000 AD	Plant propagation, nutrition, plant diseases, Kunapajala
Manasollasa	Someshvara III, 1131 AD	Gardens, crops, horticulture of the royal court
Upavana Vinoda	Sarangadhara, 13th century	Arboriculture, garden design, plant care

Q.10 Main teachings of ancient texts and their value today

contd.

प्राचीन ग्रन्थों की मुख्य शिक्षाएँ व आज का महत्त्व

Common teachings (मुख्य शिक्षाएँ)

- Classify land by fertility and water
  भूमि वर्गीकरण
- Select and treat good seed
  बीज चयन व उपचार
- Sow by season and rainfall
  ऋतु व वर्षा अनुसार बुवाई
- Use dung and organic manures
  गोबर व जैविक खाद
- Protect plants with herbal remedies
  हर्बल पौध संरक्षण
- Care for cattle - partner of farmer
  पशु देखभाल

Value today (आज का महत्त्व)

- Basis for natural and organic farming
  प्राकृतिक खेती का आधार
- Traditional weather wisdom with IMD
  परम्परागत मौसम ज्ञान
- Source of ITK for research validation
  अनुसंधान हेतु ITK
- Pride in Indian knowledge systems (NEP 2020)
  भारतीय ज्ञान परम्परा

Conclusion (निष्कर्ष)

These texts show that India had a scientific, ecology-based farming tradition thousands of years ago; their sustainable ideas are relevant even today.

ये ग्रन्थ दिखाते हैं कि भारत में हज़ारों वर्ष पहले वैज्ञानिक व पारिस्थितिक कृषि परम्परा थी।

Q.11 Describe the ancient agricultural practices of India.

10 Marks

भारत की प्राचीन कृषि पद्धतियों का वर्णन कीजिए।

Ancient Indian farmers developed practices for soil, seed, water, crops and protection that kept agriculture productive for centuries without external inputs.

प्राचीन किसानों ने मृदा, बीज, जल, फसल व संरक्षण की ऐसी पद्धतियाँ विकसित कीं जो बिना बाहरी आदानों के सदियों तक उत्पादक रहीं।

Soil and seed (मृदा व बीज)

- Land classed as urvara (fertile), usara (saline / barren)
  उर्वरा व ऊसर भूमि
- Ploughing with wooden plough and bullocks
  लकड़ी का हल
- Manuring with dung, green leaves, sheep penning
  गोबर, हरी पत्ती खाद
- Seed selected from best ears, dried and stored
  उत्तम बीज चयन
- Seed treatment with dung, honey, ghee
  बीजोपचार

Crops and water (फसल व जल)

- Mixed cropping (Kalibangan) and crop rotation
  मिश्रित फसल, फसल चक्र
- Two crops a year - kharif and rabi
  वर्ष में दो फसलें
- Wells, tanks, canals, kuhls, step wells
  कुएँ, तालाब, नहरें
- Sowing timed by nakshatra and rain
  नक्षत्र अनुसार बुवाई
- Transplanting of rice (Kashyapiya Krishisukti)
  धान रोपाई

Q.11 Plant protection, storage and animal care in ancient India

contd.

प्राचीन भारत में पौध संरक्षण, भण्डारण व पशु देखभाल

Practice	Method	Present-day form
Pest control	Neem, ash, cow urine, smoke, herbal sprays (Vrikshayurveda)	Bio-pesticides, IPM
Plant nutrition	Kunapajala, oil cakes, dung manure	Liquid organic manures, Jeevamrut
Weather forecast	Clouds, winds, stars; sayings of Ghagh - Bhaddari	Traditional + IMD agromet advisory
Storage	Mud bins, neem leaves, sun drying	Hermetic bags, scientific godowns
Animal care	Cattle as wealth; Ayurveda for animals (Shalihotra on horses)	Veterinary science, indigenous breeds
Agroforestry	Trees on bunds, sacred groves	Agroforestry policy 2014

Conclusion (निष्कर्ष)

Ancient practices were eco-friendly, low-cost and self-reliant; many are being revived under natural farming and climate-resilient agriculture.

प्राचीन पद्धतियाँ पर्यावरण-हितैषी, सस्ती व आत्मनिर्भर थीं; प्राकृतिक खेती में इनका पुनरुद्धार हो रहा है।